Grating Diffraction Diagram Generator

GUI User Guide

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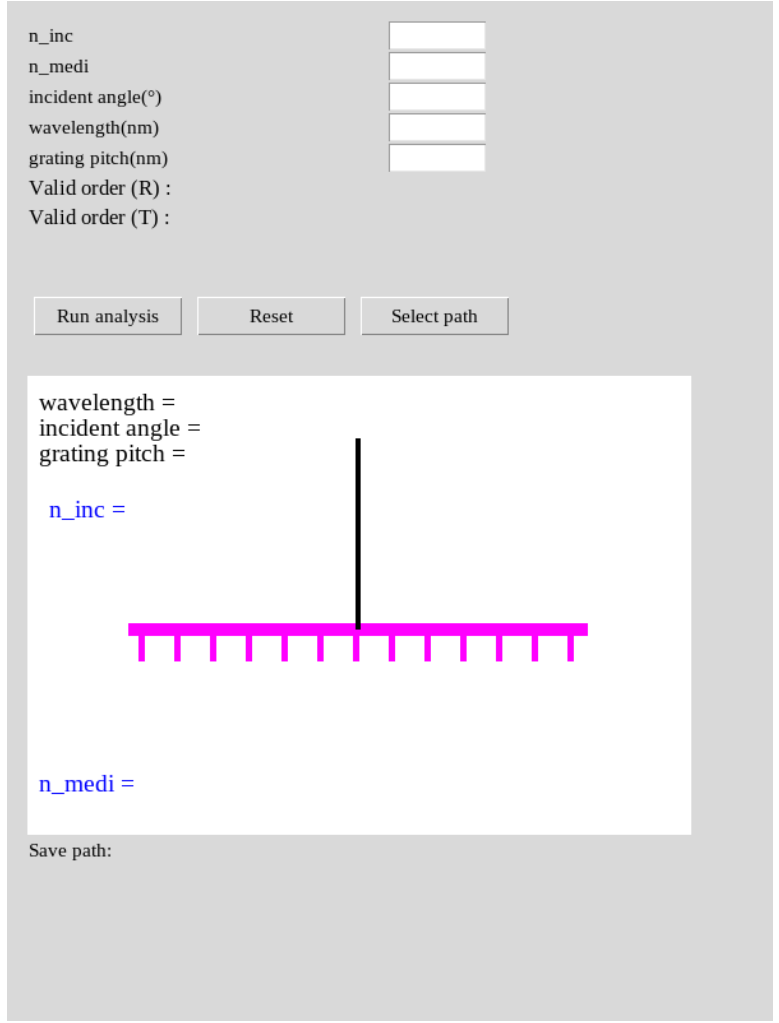
Grating Diffraction Diagram – GUI

Purpose and initial screen

Calculate reflected and transmitted diffraction angles for one wavelength.

Automatically find orders with real propagation angles.

Launch Grating Diffraction Diagram Generator.exe. All five input fields start blank.



Window title: Diffraction angle & order calculator

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1. Input parameters

GUI field	Meaning	Unit
n_inc	Refractive index of the incident-side medium	Dimensionless
n_medi	Refractive index of the transmission-side medium	Dimensionless
incident angle(°)	Signed input angle measured from the surface normal	Degrees
wavelength(nm)	Vacuum wavelength for this calculation	nm
grating pitch(nm)	Grating period	nm

Use positive indices, wavelength and pitch. Enter the incident angle between −90° and +90°.

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Grating Diffraction Diagram – GUI

2. Example inputs



Parameter	Example
n_inc / n_medi	1.0 / 1.8
Incident angle	+15°
Wavelength	532 nm
Grating pitch	700 nm

One wavelength per run. Repeat with another wavelength to compare colors.

Example values are entered manually. The tool does not preselect diffraction orders.

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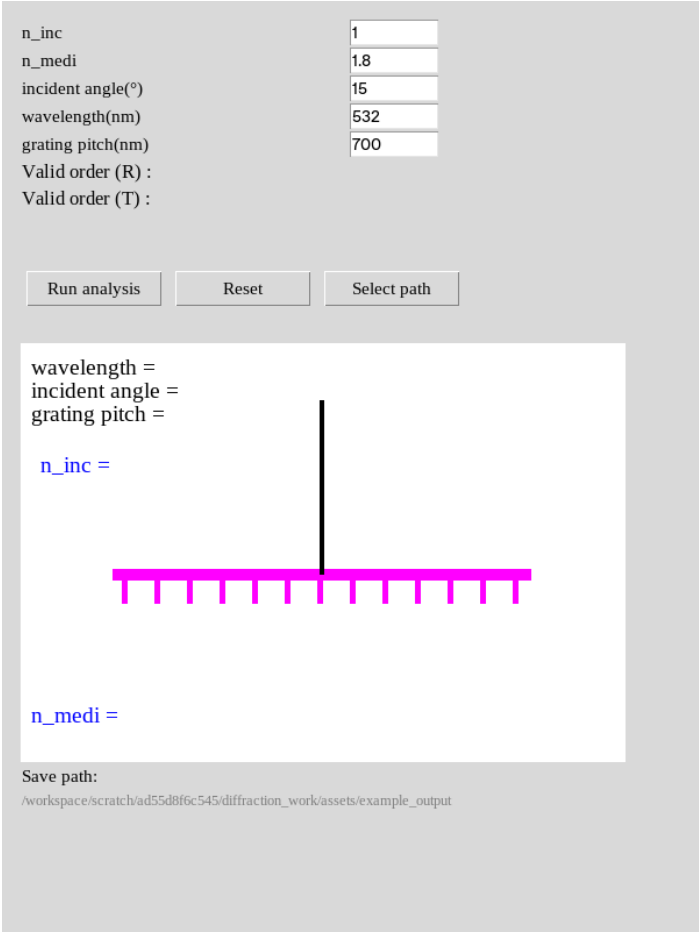
3. Select path and run

1. Fill all five numeric fields.

2. Click Select path and choose an output folder.

3. Click Run analysis.

4. Review the order lists and diagram. A successful run shows “Successfully done!”.

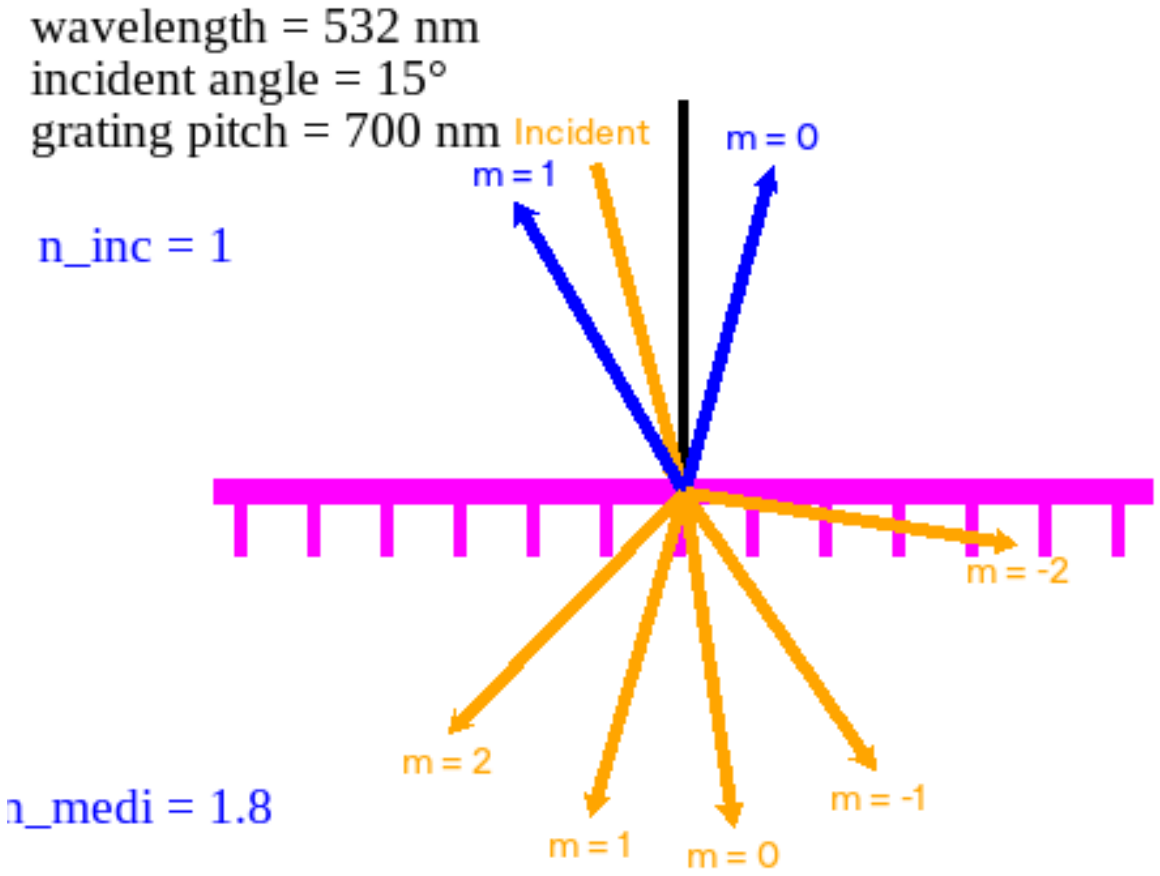


Choose a folder before running. With no selected path, this version attempts to save under D:\.

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Grating Diffraction Diagram – GUI

4. Diagram legend



Magenta: grating interface.

Black vertical line: surface normal.

Orange incoming arrow: incident beam.

Blue outgoing arrows: reflected orders (R).

Orange lower arrows: transmitted orders (T).

Labels m = … identify diffraction order. Ray length and line width do not represent optical power.

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Grating Diffraction Diagram – GUI

5. Diffraction equations and angle signs

$$q = n_{inc} \sin(\theta_{inc}) - m\lambda / \Lambda$$

$$\theta_T = \arcsin(\theta / n_{medi})$$

$$\theta_R = \arcsin(-\theta / n_{inc})$$

Λ is grating pitch. λ and Λ must use the same unit.

The GUI reports signed angles in degrees.

In this diagram, positive θ_T points right below the interface; positive θ_R points left above it.

This application uses −mλ/Λ. Match its order convention when comparing with the other GUI tools.

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6. Meaning of “Valid orders”

Test	Interpretation
Reflection: $\lvert q / n_{inc} \rvert \le 1$	A real reflected angle exists in the incident medium
Transmission: $\lvert q / n_{medi} \rvert \le 1$	A real transmitted angle exists in the second medium
Magnitude greater than 1	No real far-field propagation angle for that order
Magnitude equal to 1	A grazing direction at ±90°, included by this version

Each listed item gives the order number and angle, for example 1 : 30.08°.

An allowed direction does not imply nonzero or high diffraction efficiency.

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7. Example results

Order m	Reflection θ_R	Transmission θ_T
−2	No real angle	+81.20°
−1	No real angle	+34.47°
0	−15.00°	+8.27°
+1	+30.08°	−16.17°
+2	No real angle	−44.48°

Example: n_inc = 1, n_medi = 1.8, θ_inc = +15°, λ = 532 nm, Λ = 700 nm.

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8. Saved JPG

Run analysis automatically saves one image.

Item	Output
Parent folder	Your selected path
Application subfolder	Grating Diffraction Diagram
Run subfolder	YYYY-MM-DD_HH-MM-SS
Image filename	Diffraction_Diagram.jpg
Image dimensions	2080 × 1440 pixels

This version saves a JPG only. It does not export Excel or TXT reports.

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Grating Diffraction Diagram – GUI

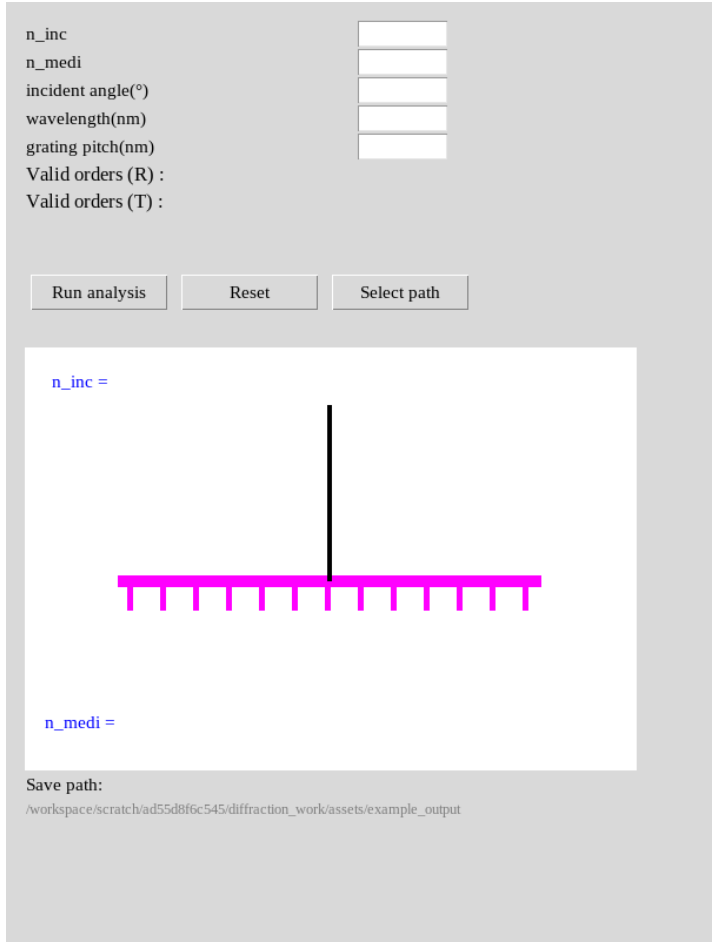
9. Reset and troubleshooting

Reset clears the five parameter fields, order lists and current rays.

The selected save path remains. Use Select path to change it.

If “Something wrong...” appears, check numeric entries, positive indices / wavelength / pitch, and folder access.

Resize the window if result text or the save-path label is cut off.



Reset does not remove previously saved images.

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Grating Diffraction Diagram – GUI

Scope and quick reference

Task	Action or scope
Generate a diagram	Enter five parameters, select a path, then run
Compare wavelengths or angles	Change the input and run each case separately
Read allowed directions	Review both R and T order lists
Assess diffraction efficiency	Requires a separate electromagnetic calculation
Check full FOV or waveguide TIR	Use the corresponding k-space / tolerance analysis tool

This tool models diffraction directions at a single grating interface.

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